Project Part 1

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```
nba <- read.csv("../data/NBA_Statistics.csv")
```

This dataset contains performance statistics from the NBA 2018-2019 season. It includes data from all 30 NBA teams. The primary focus of the dataset is on three key metrics: Win Percentage (WINP), Three Point Goal Percentage (TPGP), and Field Goal Percentage (FGP). These statistics offer insights into the effectiveness of teams in scoring and their overall performance in terms of winning games.

Here's the URL where I found my dataset: ehttps://stats.nba.com/.

My response variable is Win Percentage(Y).

My two explanatory variables are Three Point Goal Percentage and Field Goal Percentage are the predictor variables (X1 and X2, respectively).

```
nba2.lm <- lm(WINP ~ TPGP, data = nba)
summary(nba2.lm)
```

```
##
## Call:
## lm(formula = WINP ~ TPGP, data = nba)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.19906 -0.07077 -0.01591  0.05312  0.33312
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -2.26546    0.69839  -3.244 0.003047 **
## TPGP         0.07765    0.01960   3.962 0.000466 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1215 on 28 degrees of freedom
## Multiple R-squared:  0.3592, Adjusted R-squared:  0.3363
## F-statistic:  15.7 on 1 and 28 DF,  p-value: 0.0004655
```

```
# R^2 = 0.3592, Adj_r^2 = 0.3363

nba3.lm <- lm(WINP ~ FGP, data = nba)
summary(nba3.lm)
```

```
##
## Call:
## lm(formula = WINP ~ FGP, data = nba)
##
## Residuals:
```

```
##       Min        1Q    Median        3Q       Max
## -0.225929 -0.087924 -0.006762  0.092723  0.181117
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.62629    0.49619  -3.278 0.002796 **
## FGP          0.05148    0.01200   4.289 0.000193 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1179 on 28 degrees of freedom
## Multiple R-squared:  0.3965, Adjusted R-squared:  0.375
## F-statistic:  18.4 on 1 and 28 DF,  p-value: 0.0001926
```

```
#R^2 = 0.3965, Adj_r^2 = 0.375
```

For $Y \sim X1$, $r^2 = 0.3592$ and $R^2_{adj} = 0.3363$ For $Y \sim X2$ $r^2 = 0.3965$ and $R^2_{adj} = 0.375$

```
nba1.lm <- lm(WINP ~ TPGP + FGP, data = nba)
summary(nba1.lm)
```

```
##
## Call:
## lm(formula = WINP ~ TPGP + FGP, data = nba)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.24012 -0.04385 -0.00635  0.04665  0.20998
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -3.497496   0.618873  -5.651 5.33e-06 ***
## TPGP         0.062675   0.015777   3.973 0.000476 ***
## FGP          0.042741   0.009957   4.293 0.000203 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.09535 on 27 degrees of freedom
## Multiple R-squared:  0.6191, Adjusted R-squared:  0.5909
## F-statistic: 21.95 on 2 and 27 DF,  p-value: 2.189e-06
```

```
# Adj_r^2 = 0.5909
```

For $Y \sim X1 + X2$ $R^2_{adj} = 0.5909$